physics harder than the type triggered by Carl Sagan's phone call to me—thought experiments that ask, "*What things do the laws of physics permit an infinitely advanced civilization to do, and what things do the laws forbid?*" (By an "infinitely advanced civilization," I mean one whose activities are limited only by the laws of physics, and not at all by ineptness, lack of know-how, or anything else.)

We physicists, I believe, have tended to avoid such questions because they are so close to science fiction. While many of us may enjoy reading science fiction or may even write some, we fear ridicule from our colleagues for working on research close to the science fiction fringe. We therefore have tended to focus on two other, less radical, types of questions: "What kinds of things *occur naturally* in the Universe?" (for example, do black holes occur naturally? and do wormholes occur naturally?). And "What kinds of things can we as humans, with our present or near-future technology, do?" (for example, can we produce new elements such as plutonium and use them to make atomic bombs? and can we produce high-temperature superconductors and use them to lower the power bills for levitated trains and Superconducting Supercollider magnets?).

By 1988 it seemed clear to me that we physicists had been much too conservative in our questions. Already, one *Sagan-type question* (as I shall call them) was beginning to bring a payoff. By asking, "Can an infinitely advanced civilization maintain wormholes for rapid interstellar travel?" Morris and I had identified exotic material as the key to wormhole maintenance, and we had triggered a somewhat fruitful effort to understand the circumstances under which the laws of physics do and do not permit exotic material to exist.

Suppose that our Universe was created (in the big bang) with no wormholes at all. Then eons later, when intelligent life has evolved and has produced a (hypothetical) infinitely advanced civilization, *can that infinitely advanced civilization construct wormholes for rapid interstellar travel?* Do the laws of physics permit wormholes to be constructed where previously there were none? Do the laws permit this type of change in the topology of our Universe's space?

These questions are the *second half* of Carl Sagan's interstellar transport problem. The *first half*, maintaining a wormhole once it has been constructed, Sagan solved with the help of exotic matter. The second half he